Rotation-Invariant Composite Kernel Findings

21 February 2026

Questions

1. How does the 4-right-angle-averaged invariant kernel perform across rotations from 0° to 360° with a 10° step?
2. Would the same kernel perform similarly on a newly generated set of train/validation/test patches?
3. How does the invariant kernel averaged over rotations every 10° look, and how does it perform?

Rotation-Invariant Kernel Visualization

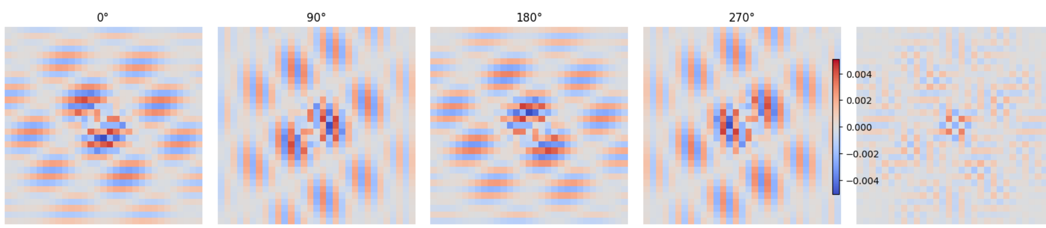

Figure 1: Invariant kernel created by averaging right rotation angles.

Data Notes

- The evaluations in this report use **newly generated patch splits** from the dataset.
- The right-angle experiment in Question 2 and the every-10°-averaged-kernel experiment in Question 3 use $N = 1750$ test samples (patches).
- In all cases, evaluation uses the kernel that was generated from the older patch set in order to check it's performance on unseen patches of data.

Results

1. 4-right-angle-averaged invariant kernel performance from 0° to 360° (step 10°)

Clarification relative to the 18 February 2026 report: In the 18 February report, the table and rotation deltas were computed only with respect to the **invariant** model at test 0°. In this 21 February report, we keep that comparison and also report deltas with respect to the original **non-invariant kernel**.

This every-10° experiment was evaluated on a **newly generated patch** of data, while using the kernel that was generated from the **older patch** set.[1]

[1]For invariant-kernel results evaluated on the same patch split that was used to generate/train the kernel (including the original composite-kernel baseline), see Table 1 in Appendix A.

Summary of every-10° results:

- Evaluated at $0^\circ, 10^\circ, \ldots, 350^\circ$ (with $360^\circ \equiv 0^\circ$).
- ACC range: 0.809091 to 0.826738; AUC range: 0.884694 to 0.898277.
- Error difference vs **invariant** 0° baseline: $\Delta E(\theta) \in [-0.017647,\ 0.000000]$.
- Error difference vs **original non-invariant** baseline: $\Delta E(\theta) \in [-0.011765,\ +0.005882]$.
- Full angle-by-angle values are listed in Appendix B.

Figure 2: Bar plot of error difference $\Delta E(\theta) = E(\theta) - E_{\text{baseline}}$ from 0° to 350°. Orange bars compare each rotated invariant-kernel result to the invariant baseline at 0°; green bars compare to the original non-invariant baseline at 0°. **Positive values mean higher error than the baseline.**

2. Performance on a newly generated patch split

Rotation-invariant kernel results:

Evaluation	Test AUC	Test ACC
Original composite kernel (non-invariant) no-rotation	0.893287	0.814973
Invariant kernel, no-rotation	0.875518	0.800571
Invariant kernel, rotated 90°	0.875518	0.800571
Invariant kernel, rotated 180°	0.875518	0.800571
Invariant kernel, rotated 270°	0.875518	0.800571

Rotation deltas (rotated − invariant kernel at test 0°):

- 90°: $\Delta\text{AUC} = +0.000000$, $\Delta\text{ACC} = +0.000000$
- 180°: $\Delta\text{AUC} = +0.000000$, $\Delta\text{ACC} = +0.000000$
- 270°: $\Delta\text{AUC} = +0.000000$, $\Delta\text{ACC} = +0.000000$

This means that the invariant kernel is stable across right-angle rotations.

Difference vs original non-invariant kernel:

$$\Delta\text{AUC} = 0.875518 - 0.893287 = -0.017769, \quad \Delta\text{ACC} = 0.800571 - 0.814973 = -0.014402$$

- AUC decrease: 0.017769.
- ACC decrease: 0.014402.
- Even with newly generated train/validation/test patches, the kernel trained on an older patch set still performs relatively close to the non-invariant baseline.

3. 10°-averaged invariant kernel: visualization and performance

In this experiment, the invariant kernel is created by averaging rotations at $0°, 10°, \ldots, 350°$, then evaluated on a newly generated split.

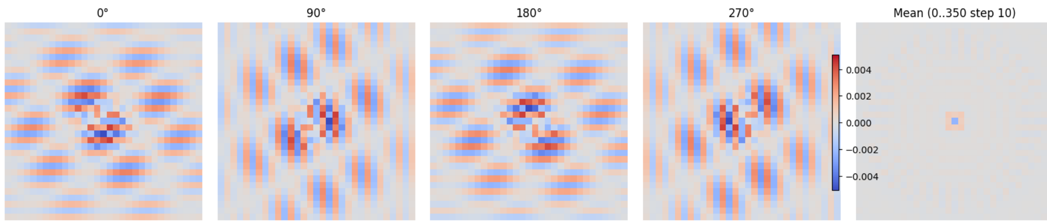

Figure 3: Invariant kernel created by averaging rotations every 10° (0° to 350°).

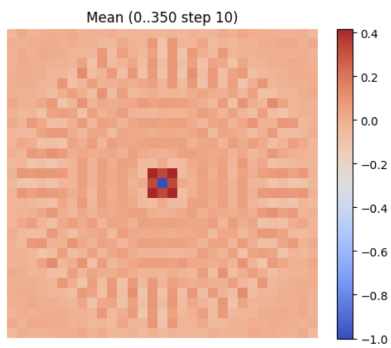

Figure 4: Additional visualization for the every-10° averaged invariant kernel.

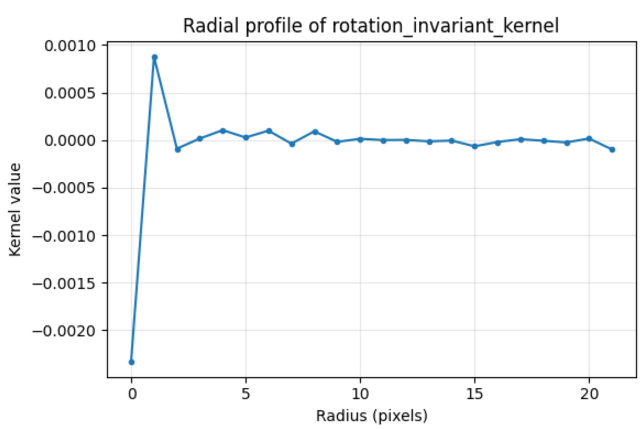

Figure 5: Radial profile of the same invariant kernel, showing mean kernel value as a function of radius from the kernel center.

Performance summary (this run):

Evaluation	Test AUC	Test ACC
Original composite kernel (non-invariant), no rotation	0.893287	0.814973
10°-averaged invariant kernel, test rot 0°	0.871238	0.798857
10°-averaged invariant kernel, test rot 90°	0.871237	0.798857
10°-averaged invariant kernel, test rot 180°	0.871237	0.798857
10°-averaged invariant kernel, test rot 270°	0.871235	0.798857

Summary points:

- Across rotations 0° to 350°, AUC ranges from 0.859495 to 0.871238 and ACC ranges from 0.782857 to 0.801714.
- Deltas vs test-rotation original (0°):

$$\Delta\mathrm{AUC} \in [-0.011743,\ 0.000000], \quad \Delta\mathrm{ACC} \in [-0.016000,\ +0.002857]$$

- Compared with the 4-right-angle-averaged kernel in Question 2 at 0° (AUC = 0.875518, ACC = 0.800571), this 10°-averaged kernel performs worse (AUC lower by 0.004280, ACC lower by 0.001714).
- Full angle-by-angle values for this 10°-averaged kernel are provided in Appendix C.

Findings

- For right-angle rotations, the invariant kernel is stable: 0°, 90°, 180°, and 270° give identical AUC/ACC in the new-split run.
- Compared with the non-invariant composite baseline, the invariant result on the new split is slightly lower (AUC by about 1.99% relative and ACC by about 1.77% relative).
- In the every-10° experiment, angle-wise behavior and error-difference trends are summarized in Question 1 and detailed in Appendix B.
- The 10°-averaged invariant kernel is also stable at right angles, but it performs worse than the 4-right-angle-averaged kernel.
- Overall, transferring the kernel to a newly generated patch split still yields relatively close performance.

Conclusion

The invariant kernel remains rotation-stable at right-angle rotations on a newly generated split, and it stays relatively close to the non-invariant composite baseline despite being transferred from an older patch set. The every-10° sweep shows consistent angle-wise behavior. The kernel averaged over every 10° performs worse than the 4-right-angle-averaged kernel in this run. Full values are reported in Appendix B and Appendix C.

A Same-Patch Right-Angle Evaluation

Setup note. The table below reports the right-angle evaluation on the same patch split used when the kernel was generated (comparable to the 18 February).

Evaluation	Test AUC	Test ACC
Original composite kernel (non-invariant)	0.893287	0.814973
Invariant kernel, test rotated 0°	0.890774	0.809091
Invariant kernel, test rotated 90°	0.890774	0.809091
Invariant kernel, test rotated 180°	0.890774	0.809091
Invariant kernel, test rotated 270°	0.890773	0.809091

Table 1: Invariant-kernel right-angle results on the same patch split used for kernel generation/training.

B Every-10° Evaluation

Setup note. The rotation-invariant kernel was created by averaging the composite kernel at 0°, 90°, 180°, and 270°. The every-10° evaluation below was run on a newly generated patch split, while using the kernel generated from the older patch set.

Detailed outputs: The complete angle-by-angle results corresponding to the main-results bar plot are listed below.

Notation note: In the raw block below, `dAUC` and `dACC` are extra right-side columns computed against the invariant-kernel average over 0°, 90°, 180°, and 270°.

```
Rotation-invariant kernel results (every 10 deg) with delta columns:
  baseline(avg 0/90/180/270) | AUC=0.890774 ACC=0.809091
  rot   0 | AUC=0.890774 ACC=0.809091 | dAUC=+0.000000 dACC=+0.000000
  rot  10 | AUC=0.884694 ACC=0.814973 | dAUC=-0.006080 dACC=+0.005882
  rot  20 | AUC=0.885241 ACC=0.818182 | dAUC=-0.005533 dACC=+0.009091
  rot  30 | AUC=0.892792 ACC=0.818717 | dAUC=+0.002018 dACC=+0.009626
  rot  40 | AUC=0.896533 ACC=0.817647 | dAUC=+0.005759 dACC=+0.008556
  rot  50 | AUC=0.893674 ACC=0.811230 | dAUC=+0.002900 dACC=+0.002139
  rot  60 | AUC=0.897000 ACC=0.826738 | dAUC=+0.006226 dACC=+0.017647
  rot  70 | AUC=0.898276 ACC=0.823529 | dAUC=+0.007502 dACC=+0.014438
  rot  80 | AUC=0.892234 ACC=0.814973 | dAUC=+0.001460 dACC=+0.005882
  rot  90 | AUC=0.890774 ACC=0.809091 | dAUC=+0.000000 dACC=+0.000000
  rot 100 | AUC=0.884694 ACC=0.814973 | dAUC=-0.006080 dACC=+0.005882
  rot 110 | AUC=0.885241 ACC=0.818182 | dAUC=-0.005533 dACC=+0.009091
  rot 120 | AUC=0.892791 ACC=0.818717 | dAUC=+0.002017 dACC=+0.009626
  rot 130 | AUC=0.896534 ACC=0.817647 | dAUC=+0.005760 dACC=+0.008556
  rot 140 | AUC=0.893675 ACC=0.811230 | dAUC=+0.002901 dACC=+0.002139
  rot 150 | AUC=0.897000 ACC=0.826738 | dAUC=+0.006226 dACC=+0.017647
  rot 160 | AUC=0.898277 ACC=0.823529 | dAUC=+0.007503 dACC=+0.014438
  rot 170 | AUC=0.892234 ACC=0.814973 | dAUC=+0.001460 dACC=+0.005882
  rot 180 | AUC=0.890774 ACC=0.809091 | dAUC=+0.000000 dACC=+0.000000
  rot 190 | AUC=0.884694 ACC=0.814973 | dAUC=-0.006080 dACC=+0.005882
  rot 200 | AUC=0.885241 ACC=0.818182 | dAUC=-0.005533 dACC=+0.009091
  rot 210 | AUC=0.892792 ACC=0.818717 | dAUC=+0.002018 dACC=+0.009626
  rot 220 | AUC=0.896533 ACC=0.817647 | dAUC=+0.005759 dACC=+0.008556
  rot 230 | AUC=0.893673 ACC=0.811230 | dAUC=+0.002899 dACC=+0.002139
  rot 240 | AUC=0.897000 ACC=0.826738 | dAUC=+0.006226 dACC=+0.017647
  rot 250 | AUC=0.898277 ACC=0.823529 | dAUC=+0.007503 dACC=+0.014438
  rot 260 | AUC=0.892234 ACC=0.814973 | dAUC=+0.001460 dACC=+0.005882
```

```
  rot 270 | AUC=0.890773 ACC=0.809091 | dAUC=-0.000001 dACC=+0.000000
  rot 280 | AUC=0.884694 ACC=0.814973 | dAUC=-0.006080 dACC=+0.005882
  rot 290 | AUC=0.885241 ACC=0.818182 | dAUC=-0.005533 dACC=+0.009091
  rot 300 | AUC=0.892792 ACC=0.818717 | dAUC=+0.002018 dACC=+0.009626
  rot 310 | AUC=0.896533 ACC=0.817647 | dAUC=+0.005759 dACC=+0.008556
  rot 320 | AUC=0.893673 ACC=0.811230 | dAUC=+0.002899 dACC=+0.002139
  rot 330 | AUC=0.897000 ACC=0.826738 | dAUC=+0.006226 dACC=+0.017647
  rot 340 | AUC=0.898276 ACC=0.823529 | dAUC=+0.007502 dACC=+0.014438
  rot 350 | AUC=0.892234 ACC=0.814973 | dAUC=+0.001460 dACC=+0.005882
```

C Every-10°-Averaged Kernel Evaluation

Setup note. In this experiment, the invariant kernel is created by averaging rotated versions of the kernel at every 10° from 0° to 350°, and then evaluating on the newly generated patch split. **Notation note.** In the block below, `dAUC` and `dACC` are `Deltas vs original` (rotation 0°), shown as right-side columns.

```
Saved composite test metrics from results.json:
  AUC=0.893287 ACC=0.814973

Rotation-invariant kernel results (every 10°) with delta columns:
  baseline(original rot 0) | AUC=0.871238 ACC=0.798857
  rot   0 | AUC=0.871238 ACC=0.798857 | dAUC=+0.000000 dACC=+0.000000
  rot  10 | AUC=0.862297 ACC=0.787429 | dAUC=-0.008941 dACC=-0.011429
  rot  20 | AUC=0.859495 ACC=0.782857 | dAUC=-0.011743 dACC=-0.016000
  rot  30 | AUC=0.868592 ACC=0.796571 | dAUC=-0.002646 dACC=-0.002286
  rot  40 | AUC=0.867827 ACC=0.801714 | dAUC=-0.003411 dACC=+0.002857
  rot  50 | AUC=0.867346 ACC=0.793714 | dAUC=-0.003892 dACC=-0.005143
  rot  60 | AUC=0.870454 ACC=0.800571 | dAUC=-0.000785 dACC=+0.001714
  rot  70 | AUC=0.863602 ACC=0.789143 | dAUC=-0.007636 dACC=-0.009714
  rot  80 | AUC=0.860216 ACC=0.789143 | dAUC=-0.011022 dACC=-0.009714
  rot  90 | AUC=0.871237 ACC=0.798857 | dAUC=-0.000001 dACC=+0.000000
  rot 100 | AUC=0.862297 ACC=0.787429 | dAUC=-0.008941 dACC=-0.011429
  rot 110 | AUC=0.859495 ACC=0.782857 | dAUC=-0.011743 dACC=-0.016000
  rot 120 | AUC=0.868590 ACC=0.796571 | dAUC=-0.002648 dACC=-0.002286
  rot 130 | AUC=0.867827 ACC=0.801714 | dAUC=-0.003411 dACC=+0.002857
  rot 140 | AUC=0.867346 ACC=0.793714 | dAUC=-0.003892 dACC=-0.005143
  rot 150 | AUC=0.870451 ACC=0.800571 | dAUC=-0.000787 dACC=+0.001714
  rot 160 | AUC=0.863606 ACC=0.789143 | dAUC=-0.007633 dACC=-0.009714
  rot 170 | AUC=0.860215 ACC=0.789143 | dAUC=-0.011023 dACC=-0.009714
  rot 180 | AUC=0.871237 ACC=0.798857 | dAUC=-0.000001 dACC=+0.000000
  rot 190 | AUC=0.862297 ACC=0.787429 | dAUC=-0.008941 dACC=-0.011429
  rot 200 | AUC=0.859495 ACC=0.782857 | dAUC=-0.011743 dACC=-0.016000
  rot 210 | AUC=0.868590 ACC=0.796571 | dAUC=-0.002648 dACC=-0.002286
  rot 220 | AUC=0.867827 ACC=0.801714 | dAUC=-0.003411 dACC=+0.002857
  rot 230 | AUC=0.867347 ACC=0.793714 | dAUC=-0.003891 dACC=-0.005143
  rot 240 | AUC=0.870452 ACC=0.800571 | dAUC=-0.000786 dACC=+0.001714
  rot 250 | AUC=0.863602 ACC=0.789143 | dAUC=-0.007636 dACC=-0.009714
  rot 260 | AUC=0.860216 ACC=0.789143 | dAUC=-0.011022 dACC=-0.009714
  rot 270 | AUC=0.871235 ACC=0.798857 | dAUC=-0.000003 dACC=+0.000000
  rot 280 | AUC=0.862297 ACC=0.787429 | dAUC=-0.008941 dACC=-0.011429
  rot 290 | AUC=0.859495 ACC=0.782857 | dAUC=-0.011743 dACC=-0.016000
  rot 300 | AUC=0.868590 ACC=0.796571 | dAUC=-0.002648 dACC=-0.002286
  rot 310 | AUC=0.867827 ACC=0.801714 | dAUC=-0.003411 dACC=+0.002857
  rot 320 | AUC=0.867346 ACC=0.793714 | dAUC=-0.003892 dACC=-0.005143
```

```
rot 330 | AUC=0.870452 ACC=0.800571 | dAUC=-0.000787 dACC=+0.001714
rot 340 | AUC=0.863602 ACC=0.789143 | dAUC=-0.007636 dACC=-0.009714
rot 350 | AUC=0.860216 ACC=0.789143 | dAUC=-0.011022 dACC=-0.009714
```